

Item	Spec
Detection algorithm	2-state Gaussian HMM, fit weekly on rolling 250 days
Input features	[Nifty 5d log-return, India VIX level, Nifty-500 % above 50-DMA]
Output	{regime, P(BULL), P(BEAR), P(SIDEWAYS), confidence}
Dwell time	3 consecutive daily reads in new regime before switching
Hysteresis	Enter BEAR at $P(\text{BEAR}) > 0.70$; exit BEAR at $P(\text{BULL}) > 0.55$ (asymmetric — easier to leave defensive)
Cooldown	After any flip, lock regime for 6 bars (3 hours)
TRANSITIONING band	When $0.50 < P(\text{new}) < 0.70$: keep previous regime, allocation $\times 0.5$
Intraday confirm	Kaufman Efficiency Ratio on 5-min Nifty modulates <i>size only</i> , never flips daily regime
Skip opening 30 min	First read at 10:00 IST
Look-ahead bug	Fix <code>_load_csv</code> to use point-in-time slicing

Why this fixes 04-29: On a +0.81% Nifty day with breadth >55%, the HMM posterior $P(\text{BEAR})$ would have been $\approx 0.40\text{--}0.55$ — *not* >0.70. The TRANSITIONING band would have triggered: previous regime kept (whatever it was the day before), allocation halved. The 175 blocked LONGs would have run at half size instead of being blocked entirely.

Phase 2 – Regime as a feature (2 weeks)

Goal: ingest regime probability into the v4 scorer.